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Communication control method and user equipment for transitioning between RRC modes

Abstract

A communication control method comprises transmitting, by a user equipment to a network node, first information indicating that the user equipment prefers one of a RRC idle mode and a RRC inactive mode when data to be transmitted to the network node is not to occur in near future; and in response to changing a preferred RRC mode from a RRC mode indicated by the first information to a RRC connected mode, transmitting, by the user equipment to the network node, second information indicating that the user equipment prefers to revert the first information to leave the RRC connected mode.

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Background/Summary

RELATED APPLICATIONS (1) The present application is a Continuation of U.S. patent application Ser. No. 17/456,620, filed on Nov. 26, 2021, which is a Continuation based on PCT Application No. PCT/JP2020/020017, filed on May 20, 2020, which claims the benefit of Japanese

TECHNICAL FIELD

(1) The present disclosure relates to a communication control method and a user equipment in mobile communication systems.

BACKGROUND ART

(2) The 3rd Generation Partnership Project (3GPP), a standardization project for mobile communication systems, has defined, as modes of Radio Resource Control (RRC) for a user equipment, an RRC connected mode, an RRC inactive mode, and an RRC idle mode.

(3) The RRC connected mode and the RRC inactive mode are modes in which an RRC connection for a user equipment is established. The RRC inactive mode is a mode in which an established RRC connection is suspended. The RRC idle mode is a mode in which an RRC connection for the user equipment is not established.

(4) The user equipment in the RRC idle mode or the RRC inactive mode needs to monitor a downlink control channel only in a periodic paging occasion, and thus, power consumption of the user equipment is small. On the other hand, the user equipment in the RRC connected mode needs to frequently monitor at least the downlink control channel in order to perform data communication, and the power consumption of the user equipment is large.

(5) Thus, there is a demand to realize a technique in which a user equipment can appropriately transition from an RRC connected mode to an RRC idle mode or an RRC inactive mode in order to reduce the power consumption of the user equipment.

SUMMARY

(6) A communication control method according to an embodiment comprises transmitting, by a user equipment to a network node, first information indicating that the user equipment prefers one of a RRC idle mode and a RRC inactive mode when data to be transmitted to the network node is not to occur in near future; and in response to changing a preferred RRC mode from a RRC mode indicated by the first information to a RRC connected mode, transmitting, by the user equipment to the network node, second information indicating that the user equipment prefers to revert the first information to leave the RRC connected mode.

(7) A user equipment according to an embodiment comprises a controller configured to transmit, to a network node, first information indicating that the user equipment prefers one of a RRC idle mode and a RRC inactive mode when data to be transmitted to the network node is not to occur in near future; and in response to changing a preferred RRC mode from a RRC mode indicated by the first information to a RRC connected mode, transmit, to the network node, second information indicating that the user equipment prefers to revert the first information to leave the RRC connected mode.

(8) A chipset according to an embodiment for a user equipment is configured to execute processing of transmitting, to a network node, first information indicating that the user equipment prefers one of a RRC idle mode and a RRC inactive mode when data to be transmitted to the network node is not to occur in near future; and in response to changing a preferred RRC mode from a RRC mode indicated by the first information to a RRC connected mode, transmitting, to the network node, second information indicating that the user equipment prefers to revert the first information to leave the RRC connected mode.

(9) A non-transitory computer-readable medium according to an embodiment comprises, stored thereupon, computer program instructions for execution by a user equipment. The program instructions are configured to cause the user equipment to execute processing of transmitting, to a network node, first information indicating that the user equipment prefers one of a RRC idle mode and a RRC inactive mode when data to be transmitted to the network node is not to occur in near future; and in response to changing a preferred RRC mode from a RRC mode indicated by the first

information to a RRC connected mode, transmitting, to the network node, second information indicating that the user equipment prefers to revert the first information to leave the RRC connected mode.

(10) A system according to an embodiment comprises a user equipment configured to transmit, to a network node, first information indicating that the user equipment prefers one of a RRC idle mode and a RRC inactive mode when data to be transmitted to the network node is not to occur in near future; and in response to changing a preferred RRC mode from a RRC mode indicated by the first information to a RRC connected mode, transmit, to the network node, second information indicating that the user equipment prefers to revert the first information to leave the RRC connected mode.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. 1 is a diagram illustrating a configuration of a mobile communication system according to an embodiment.

(2) FIG. 2 is a diagram illustrating a configuration of a user equipment according to an embodiment.

(3) FIG. 3 is a diagram illustrating a configuration of a base station according to the embodiment.

(4) FIG. 4 is a diagram illustrating a configuration of a radio interface protocol stack in a user plane according to an embodiment.

(5) FIG. 5 is a diagram illustrating a configuration of a radio interface protocol stack in a control plane according to an embodiment.

(6) FIG. 6 is a diagram illustrating basic operations related to a transition from an RRC connected mode to another mode.

(7) FIG. 7 is a diagram illustrating operations of the mobile communication system according to a first embodiment.

(8) FIG. 8 is a diagram illustrating operations of the mobile communication system according to modification 1 of the first embodiment.

(9) FIG. 9 is a diagram illustrating operations of the mobile communication system according to modification 2 of the first embodiment.

(10) FIG. 10 is a diagram illustrating operations of the mobile communication system according to a second embodiment.

(11) FIG. 11 is a diagram illustrating operations according to other embodiments.

DESCRIPTION OF EMBODIMENTS

(12) A mobile communication system according to an embodiment will be described with reference to the drawings. In the description of the drawings, the same or similar parts are designated with the same or similar reference signs.

(13) Mobile Communication System

(14) First, a configuration of a mobile communication system according to an embodiment will be described. Although the mobile communication system according to the embodiment is a 5G system of 3GPP, LTE may be at least partially applied to the mobile communication system.

(15) FIG. 1 is a diagram illustrating a configuration of the mobile communication system according to the embodiment.

(16) As illustrated in FIG. 1, the mobile communication system includes a User Equipment (UE) **100**, a 5G radio access network (Next Generation Radio Access Network (NG-RAN)) **10**, and a 5G core network (5GC) **20**.

(17) The UE **100** is a movable apparatus. The UE **100** may be any apparatus so long as it is an apparatus utilized by a user. Examples of the UE **100** include a mobile phone terminal (including a

smartphone), a tablet terminal, a laptop, a communication module (including a communication card or a chipset), a sensor or an apparatus provided in a sensor, a vehicle or an apparatus provided in a vehicle (vehicle UE), or an air vehicle or an apparatus provided in an air vehicle (aerial UE).

(18) The NG-RAN **10** includes base stations (each of which is referred to as a “gNB” in the 5G system) **200**. The gNB **200** may be also referred to as an NG-RAN node. The gNBs **200** are connected to each other via an Xn interface that is an inter-base station interface. The gNB **200** manages one or more cells. The gNB **200** performs radio communication with the UE **100** that has established a connection with a cell of the gNB **200**. The gNB **200** has a radio resource management (RRM) function, a user data (hereinafter simply referred to as “data”) routing function, a measurement control function for mobility control and scheduling, and/or the like. A “cell” is used as a term that indicates a minimum unit of a radio communication area. A “cell” is also used as a term that indicates a function or resource that performs radio communication with the UE **100**. One cell belongs to one carrier frequency.

(19) Note that the gNB may be connected to an Evolved Packet Core (EPC) which is an LTE core network, or an LTE base station may be connected to a 5GC. Moreover, the LTE base station may be connected to the gNB via the inter-base station interface.

(20) The 5GC **20** includes an Access and Mobility Management Function (AMF) and a User Plane Function (UPF) **300**. The AMF performs various types of mobility control for the UE **100**, and the like. The AMF manages information of an area in which the UE **100** exists by communicating with the UE **100** by using Non-Access Stratum (NAS) signaling. The UPF performs data transfer control. The AMF and the UPF are connected to the gNB **200** via an NG interface which is a base station to core network interface.

(21) FIG. **2** is a diagram illustrating a configuration of the UE **100** (user equipment).

(22) As illustrated in FIG. **2**, the UE **100** includes a receiver **110**, a transmitter **120**, and a controller **130**.

(23) The receiver **110** performs various types of reception under control of the controller **130**. The receiver **110** includes an antenna and a receiving unit. The receiving unit converts a radio signal received by the antenna into a baseband signal (reception signal) and outputs the signal to the controller **130**.

(24) The transmitter **120** performs various type of transmission under control of the controller **130**. The transmitter **120** includes the antenna and a transmitting unit. The transmitting unit converts the baseband signal (transmission signal) output by the controller **130** into a radio signal and transmits the signal from the antenna.

(25) The controller **130** performs various type of control in the UE **100**. The controller **130** includes at least one processor and at least one memory electrically connected to the processor. The memory stores programs to be executed by the processor and information used for processing by the processor. The processor may include a baseband processor and a Central Processing Unit (CPU). The baseband processor performs modulation/demodulation and coding/decoding of the baseband signal, and the like. The CPU executes the programs stored in the memory to perform various types of process.

(26) FIG. **3** is a diagram illustrating a configuration of the gNB **200** (base station).

(27) As illustrated in FIG. **3**, the gNB **200** includes a transmitter **210**, a receiver **220**, a controller **230**, and a backhaul communicator **240**.

(28) The transmitter **210** performs various types of transmission under control of the controller **230**. The transmitter **210** includes an antenna and a transmitting unit. The transmitting unit converts a baseband signal (transmission signal) output by the controller **230** into a radio signal and transmits the signal from the antenna.

(29) The receiver **220** performs various types of reception under control of the controller **230**. The receiver **220** includes the antenna and a receiving unit. The receiving unit converts the radio signal received by the antenna into a baseband signal (reception signal) and outputs the signal to the

controller **230**.

(30) The controller **230** performs various type of control in the gNB **200**. The controller **230** includes at least one processor and at least one memory electrically connected to the processor. The memory stores programs to be executed by the processor and information used for processing by the processor. The processor may include a baseband processor and a CPU. The baseband processor performs modulation/demodulation and coding/decoding of the baseband signal, and the like. The CPU executes the programs stored in the memory to perform various types of process.

(31) The backhaul communicator **240** is connected to a neighboring base station via the inter-base station interface. The backhaul communicator **240** is connected to the AMF/UPF **300** via the base station to core network interface. Note that the gNB may include a Central Unit (CU) and a Distributed Unit (DU) (i.e., may be functionally divided), and both units may be connected to each other via an F1 interface.

(32) FIG. **4** is a diagram illustrating a configuration of a radio interface protocol stack in a user plane handling data.

(33) As illustrated in FIG. **4**, the radio interface protocol in the user plane includes a physical (PHY) layer, a Medium Access Control (MAC) layer, a Radio Link Control (RLC) layer, a Packet Data Convergence Protocol (PDCP) layer, and a Service Data Adaptation Protocol (SDAP) layer.

(34) The PHY layer performs coding/decoding, modulation/demodulation, antenna mapping/demapping, and resource mapping/demapping. Data and control information are transmitted via a physical channel between the PHY layer of the UE **100** and the PHY layer of the gNB **200**.

(35) The MAC layer performs priority control of data, retransmission processing by hybrid ARQ (HARQ), a random access procedure, and the like. Data and control information are transmitted via a transport channel between the MAC layer of the UE **100** and the MAC layer of the gNB **200**. The MAC layer of the gNB **200** includes a scheduler. The scheduler determines uplink and downlink transport formats (a transport block size, and a modulation and coding scheme (MCS)) and resource blocks allocated to the UE **100**.

(36) The RLC layer transmits data to the RLC layer on the receiver side using the functions of the MAC layer and PHY layer. Data and control information are transmitted between the RLC layer of the UE **100** and the RLC layer of the gNB **200** via a logical channel.

(37) The PDCP layer performs header compression/extension and encryption/decryption.

(38) The SDAP layer performs mapping between an IP flow that is a unit by which the core network performs QoS control and a radio bearer that is a unit by which an Access Stratum (AS) performs QoS control. Note that in a case where a RAN is connected to the EPC, SDAP is not necessary.

(39) FIG. **5** is a diagram illustrating a configuration of a radio interface protocol stack in a control plane handling signaling (control signal).

(40) As illustrated in FIG. **5**, the radio interface protocol stack in the control plane includes a Radio Resource Control (RRC) layer and a Non-Access Stratum (NAS) layer instead of the SDAP layer illustrated in FIG. **4**.

(41) RRC signaling for various types of configuration is transmitted between the RRC layer of the UE **100** and the RRC layer of the gNB **200**. The RRC layer controls the logical channel, the transport channel, and the physical channel in response to establishing, re-establishing, and releasing the radio bearer. In a case where there is a connection (RRC connection) between the RRC of the UE **100** and the RRC of the gNB **200**, the UE **100** is in an RRC connected mode. In a case where there is no connection (RRC connection) between the RRC of the UE **100** and the RRC of the gNB **200**, the UE **100** is in an RRC idle mode. In a case where the RRC connection is suspended, the UE **100** is in an RRC inactive mode.

(42) The NAS layer located higher than the RRC layer performs session management, mobility management, and the like. NAS signaling is transmitted between the NAS layer of the UE **100** and

the NAS layer of the AMF **300**.

(43) Note that the UE **100** includes an application layer or the like other than the radio interface protocol.

(44) Transition from RRC Connected Mode to Another Mode

(45) Next, a transition from the RRC connected mode to another mode will be described.

(46) The RRC connected mode and the RRC inactive mode are modes in which an RRC connection for the UE **100** is established. However, the RRC inactive mode is a mode in which an established RRC connection is suspended. Specifically, in the RRC inactive mode, context information for the UE **100** is retained in the gNB **200** and the UE **100**, and thus, the UE **100** can smoothly transition to the RRC connected mode by using the retained context information. The RRC idle mode is a mode in which an RRC connection for the UE **100** is not established.

(47) The UE **100** in the RRC idle mode or the RRC inactive mode needs to monitor a downlink control channel only in a periodic paging occasion, and thus, power consumption of the UE **100** is small. On the other hand, the UE **100** in the RRC connected mode needs to frequently monitor at least the downlink control channel in order to perform data communication, and the power consumption of the UE **100** is large. Regarding the uplink as well, the UE **100** in the RRC connected mode needs to periodically perform transmission of an uplink control channel (PUCCH), that is, Channel State Information (CSI) feedback, and the power consumption of the UE **100** is large.

(48) FIG. **6** is a diagram illustrating basic operations related to the transition from the RRC connected mode to another mode.

(49) As illustrated in FIG. **6**, in step **S1**, the UE **100** is in the RRC connected mode in a cell of the gNB **200**. The UE **100** in the RRC connected mode performs data communication with the gNB **200**.

(50) In step **S2**, the UE **100** transmits uplink data to the gNB **200** via a Physical Uplink Shared Channel (PUSCH), and receives downlink data from the gNB **200** via a Physical Downlink Shared Channel (PDSCH).

(51) In step **S3**, the UE **100** and the gNB **200** complete the data communication. Completion of the data communication refers to uplink data transmission completion when only uplink data transmission is performed, refers to downlink data transmission completion when only downlink data transmission is performed, and refers to uplink and downlink data transmission completion when uplink and downlink data transmission is performed.

(52) In step **S4**, the UE **100** may transmit, to the gNB **200**, a Release Assistance Indicator (RAI) which is an indicator indicating that data to be transmitted to the gNB **200** will not occur in the near future. The RAI may be a buffer status report indicating that a buffer size value is zero. The RAI may be enabled only when RAI activation is configured from the gNB **200**. However, step **S4** is not necessarily required and step **S5** may be performed without step **S4**.

(53) In step **S5**, the gNB **200** transmits, to the UE **100**, an RRC release message which is a type of dedicated RRC message that is transmitted by UE-specific unicast transmission, in response to completion of data communication. In a case where the gNB **200** determines to make the UE **100** transition to the RRC inactive mode, the gNB **200** includes configuration information for the RRC inactive mode (SuspendConfig) in the RRC release message. SuspendConfig includes *ran-PagingCycle* which is a discontinuous reception (DRX) cycle for the RRC inactive mode, and *fullI-RNTI* or *shortI-RNTI* which is an identifier assigned to the UE **100** for the RRC inactive mode. On the other hand, in a case where the gNB **200** determines to make the UE **100** transition to the RRC idle mode, the gNB **200** does not include SuspendConfig in the RRC release message.

(54) In step **S6**, the UE **100** transitions to the RRC idle mode or the RRC inactive mode, based on the RRC release messages received from the gNB **200**. Specifically, the UE **100** transitions to the RRC inactive mode in a case where the RRC release message includes SuspendConfig, or the RRC idle mode in a case where the RRC release message does not include SuspendConfig.

(55) Here, the UE **100** (RRC layer) transitions to the RRC idle mode or the RRC inactive mode at a timing that is the earlier one of a timing at which 60 ms elapses from receiving the RRC message and a timing at which a lower layer (layer 2) sends an acknowledgment in response to the RRC message.

(56) In the basic operations illustrated in FIG. 6, the UE **100** needs to maintain the RRC connected mode despite the data communication completion, for a period T1 from completing the data communication in step S3 to transitioning to the RRC idle mode or the RRC inactive mode in step S6. The power of the UE **100** consumed during the period T1 is wasteful power that does not contribute to transmission and reception of data, and thus it is considered that there is room for reduction of the power consumption of UE **100** by reducing the period T1.

(57) It can be also thought that in a case where, after a long time elapses from the completion of data communication (step S3), the next data communication is initiated, making the UE **100** transition to the RRC idle mode preferable; otherwise, making the UE **100** transition to the RRC inactive mode preferable. However, the gNB **200** may not have sufficient criteria to determine whether to make the UE **100** transition to the RRC idle mode or the RRC inactive mode.

First Embodiment

(58) Next, a first embodiment will be described. FIG. 7 is a diagram illustrating operations of the mobile communication system according to the first embodiment.

(59) As illustrated in FIG. 7, in step S101, the UE **100** is in the RRC connected mode in a cell of the gNB **200**. The UE **100** in the RRC connected mode performs data communication with the gNB **200**.

(60) In step S102, the gNB **200** transmits, to the UE **100**, an RRC release message with a condition for releasing or suspending the RRC connection of the UE **100** (hereinafter referred to as a “predetermined condition”). Such a conditional RRC release message may be an RRC release message including condition information specifying the predetermined condition. The conditional RRC release message can be a new RRC release message having a different format from the general RRC release message.

(61) The UE **100** immediately transitions to the RRC idle mode or the RRC inactive mode when the UE **100** receives a general RRC release message. On the other hand, when the UE **100** receives the conditional RRC release message, the UE **100** leaves the transition to the RRC idle mode or RRC inactive mode pending, and maintains the RRC connected mode until the predetermined condition is met.

(62) The predetermined condition may be configured for the UE **100** as configuration information (information element) in the RRC release message, or a predetermined condition defined by communication standards may be configured for the UE **100** in advance at the time of shipping the UE **100**. Details of the predetermined condition are described in step S104.

(63) In a case where the gNB **200** determines to make the UE **100** transition to the RRC inactive mode, the gNB **200** includes configuration information for the RRC inactive mode (SuspendConfig) in the RRC release message. On the other hand, in a case where the gNB **200** determines to make the UE **100** transition to the RRC idle mode, the gNB **200** does not include SuspendConfig in the RRC release message.

(64) The UE **100** continuously checks whether the predetermined condition is met in response to reception of a conditional RRC release message from the gNB **200**.

(65) In step S103, the UE **100**, after receiving the conditional RRC release message, performs data communication with the gNB **200** until the predetermined condition is met. For example, the UE **100** transmits uplink data to the gNB **200** via the PUSCH, or receives downlink data from the gNB **200** via the PDSCH.

(66) In step S104, the UE **100** determines that the predetermined condition has been met. The predetermined condition is any one of conditions 1 to 3 below, or a combination of two or more of the conditions. In a case where the gNB **200** configures a predetermined condition for the UE **100**,

the gNB **200** may include one or more identifiers of the conditions 1 to 3 in the conditional RRC release message.

(67) (1) Condition 1: The UE **100** completes the uplink data transmission to the gNB **200** (i.e., the last uplink data transmission ends).

(68) For example, in a situation in which the gNB **200** has completed transmission of downlink data to the UE **100** and receives uplink data from the UE **100**, the gNB **200** transmits an RRC release message specifying the condition 1 to the UE **100**. In this case, the UE **100** is to autonomously transition to the RRC idle mode or the RRC inactive mode upon completion of the uplink transmission.

(69) The completion of the uplink data transmission in the condition 1 may be a condition 1A that the UE **100** transmits, to the gNB **200**, information indicating completion of the data transmission. For example, the UE **100** transmits the RAI described above to the gNB **200** as a complete notification (end marker) of the uplink data transmission, and autonomously transitions to the RRC idle mode or the RRC inactive mode at the time point of the end marker transmission.

(70) The completion of the uplink data transmission in the condition 1 may be a condition 1B that the data transmission is completed for an amount of data or a transmission time that is notified in advance between the UE **100** and the gNB **200** before the completion of the data transmission. For example, prior to step S104, the UE **100** notifies the gNB **200** in advance of the amount of uplink data to be transmitted to the gNB **200** or the time to be taken to complete the uplink data transmission to the gNB **200**. Such notifications may be made by way of the RAI. Then, the UE **100** autonomously transitions to the RRC idle mode or the RRC inactive mode when completing the data transmission of the amount of data notified in advance to the gNB **200** or the data transmission of the time notified in advance to the gNB **200**.

(71) The completion of the uplink data transmission in the condition 1 may be a condition 1C that, after the UE **100** has performed the last uplink data transmission, an acknowledgment (HARQ ACK or ARQ ACK) in response to this uplink data is received from the gNB **200**. When receiving the acknowledgment corresponding to the last uplink data transmission from the gNB **200**, the UE **100** autonomously transitions to the RRC idle mode or the RRC inactive mode. Note that, at the time of the last uplink data transmission, the UE **100** may transmit the end marker to the gNB **200** in order to indicate to the gNB **200** that it is the last uplink data transmission. For example, the UE **100** may transmit the MAC CE corresponding to the end marker to the gNB **200** with the same MAC PDU as the last uplink data.

(72) The completion of the uplink data transmission under the condition 1 may be a combination of two or more of the conditions 1A to 1C.

(73) (2) Condition 2: The UE **100** completes the downlink data reception from the gNB **200** (i.e., the last downlink data reception ends).

(74) For example, in a situation in which the gNB **200** has completed reception of uplink data from the UE **100** and transmits downlink data to the UE **100**, the gNB **200** transmits an RRC release message specifying the condition 2 to the UE **100**. In this case, the UE **100** is to autonomously transition to the RRC idle mode or the RRC inactive mode upon completing the downlink data reception.

(75) The completion of the downlink data reception in the condition 2 may be a condition 2A that the UE **100** receives information from the gNB **200** indicating the completion of the data transmission. For example, the gNB **200** transmits a complete notification (end marker) of the last downlink data transmission to the UE **100**, and the UE **100** autonomously transitions to the RRC idle mode or the RRC inactive mode at the time point of the end marker reception. Note that, at the time of the last downlink data transmission, the gNB **200** may transmit the end marker to the UE **100** in order to indicate to the UE **100** that it is the last downlink data transmission. For example, the gNB **200** may transmit the MAC CE corresponding to the end marker to the UE **100** with the same MAC PDU as the last downlink data.

(76) The completion of the downlink data reception in the condition 2 may be a condition 2B that the data reception is completed for an amount of data or a reception time that is notified in advance to the UE **100** by the gNB **200** before the completion of the downlink data reception. For example, prior to step **S104**, the gNB **200** notifies the UE **100** in advance of the amount of downlink data to be transmitted to the UE **100** or the time to be taken to complete the downlink data transmission to the UE **100**. Then, the UE **100** autonomously transitions to the RRC idle mode or the RRC inactive mode upon completion of reception of the amount of data notified in advance from the gNB **200** or the reception in the time notified in advance from the gNB **200**.

(77) The completion of the downlink data reception in the condition 2 may be a condition 2C that after the UE **100** has performed the last downlink data reception, the UE **100** has transmitted to the gNB **200**, an acknowledgment (HARQ ACK or ARQ ACK) in response to this downlink data. When transmitting, to the gNB **200**, the acknowledgment corresponding to the last downlink data transmission, the UE **100** autonomously transitions to the RRC idle mode or the RRC inactive mode.

(78) The completion of the downlink data reception in the condition 2 may be a combination of two or more of the conditions 2A to 2C.

(79) (3) Condition 3: The time specified by way of the RRC release message from the gNB **200** elapses.

(80) For example, the gNB **200** includes a timer value that defines a time for which the UE **100** is made to maintain the RRC connected mode in a conditional RRC release message. The UE **100** starts the timer with the timer value being configured when receiving the RRC release message and autonomously transitions to the RRC idle mode or the RRC inactive mode when the timer expires.

(81) In step **S105**, the UE **100** transitions to the RRC idle mode or the RRC inactive mode in response to the predetermined condition being met. The UE **100** transitions to the RRC inactive mode in a case where the RRC release message received in step **S102** includes SuspendConfig, or the RRC idle mode in a case where the RRC release message does not include SuspendConfig.

(82) On the other hand, in step **S106**, the gNB **200** determines that the predetermined condition is met and the UE **100** assumes that the UE **100** transitions to a mode selected by the gNB **200** of the RRC idle mode or the RRC inactive mode. Then, the gNB **200** notifies the core network (AMF **300**) of the mode of the UE **100** after the transition and an identifier of the UE **100**. The notification may be performed on the NG interface.

(83) According to the first embodiment, a period T2 from completing the data communication in step **S104** to transitioning to the RRC idle mode or the RRC inactive mode in step **S105** can be shortened. Specifically, the period T2 illustrated in FIG. 7 is shorter than the period T1 illustrated in FIG. 6. Thus, the power consumption of the UE **100** can be reduced compared to the basic operations illustrated in FIG. 7.

Modification 1 of First Embodiment

(84) The above embodiment describes an example in which the UE **100** transmits the RAI to the gNB **200** before the gNB **200** transmits the conditional RRC release message to the UE **100**.

(85) In the present modification, a specific example of such an RAI will be described. FIG. 8 is a diagram illustrating operations according to the present modification. As illustrated in FIG. 8, in step **S107**, the UE **100** transmits the RAI to the gNB **200** before the gNB **200** transmits the conditional RRC release message to the UE **100**. The RAI may be a type of an RRC message. The UE **100** includes, in the RAI, at least one piece of information (information element) out of following “information 1” to “information 4”. Information 1: information indicating a mode desired by the UE **100** out of the RRC idle mode and the RRC inactive mode.

(86) For example, the UE **100** predicts a length of a period in which uplink data transmitted to the gNB **200** and/or downlink data received from the gNB **200** is not generated. When the period is a certain period or less, the UE **100** selects the RRC inactive mode, and otherwise, the UE **100** selects the RRC idle mode, and the UE **100** includes the information indicating the selected mode

in the RAI. Information 2: Auxiliary information for the gNB **200** to determine a condition configuration value when the predetermined condition (trigger condition) as described above has been determined.

(87) For example, when condition 3 described above, in other words, the condition that the time (timer value) specified by way of the conditional RRC release message from the gNB **200** has elapsed, is used as the trigger condition, the UE **100** includes the auxiliary information for the gNB **200** to determine the timer value in the RAI. The UE **100** may predict in how much more time the data communication ends, and include the predicted time in the RAI. The gNB **200** determines the timer value included in the conditional RRC release message, based on the time notified from the UE **100**. In this manner, for example, the UE **100** can transition to the RRC idle mode or the RRC inactive mode without waiting for confirmation of the HARQ ACK or the RLC ACK, which has hitherto been necessary in RRC Release, and thus time from completion of data communication to transition to the RRC idle mode or the RRC inactive mode can be reduced.

(88) Alternatively, when a condition that radio quality (RSRP, RSRQ) has fallen below a threshold is used as the trigger condition (hereinafter referred to as “condition 4”), the UE **100** includes the auxiliary information for the gNB **200** to determine the threshold in the RAI. For example, the UE **100** may identify the minimum radio quality necessary for satisfying required QoS according to a type of an application executed by the UE **100**, and include a value of the identified radio quality in the RAI. The gNB **200** determines a radio quality threshold included in the conditional RRC release message, based on the value of the radio quality notified from the UE **100**.

(89) Alternatively, when a condition that the amount of data in a buffer of the UE **100** has fallen below a threshold is used as the trigger condition (hereinafter referred to as “condition 5”), the UE **100** includes the auxiliary information for the gNB **200** to determine the threshold in the RAI.

Information 3: Auxiliary information for the gNB **200** to determine which condition is configured by the gNB **200** in the conditional RRC release message.

(90) The UE **100** includes the auxiliary information for the gNB **200** to determine any one of conditions 1 to 5 described above in the RAI. For example, the UE **100** includes a condition identifier indicating a condition desired by the UE **100**, out of conditions 1 to 5, in the RAI. When the UE **100** executes such an application that the UE **100** performs communication only when the radio quality is satisfactory, the UE **100** may include a condition identifier indicating condition 4 as the desired condition in the RAI. Information 4: Auxiliary information for the gNB **200** to determine which of the conditional RRC release message or a regular RRC release message is transmitted.

(91) The UE **100** includes information indicating whether the UE **100** desires the gNB **200** to transmit the conditional RRC release message or desires the gNB **200** to transmit the regular RRC release message (specifically, an unconditional RRC release message) in the RAI.

(92) Note that the RAI according to the present modification may implicitly indicate, for the gNB **200**, whether or not the UE **100** supports reception of the conditional RRC release message.

Specifically, only the UE **100** that supports reception of the conditional RRC release message can be configured to transmit the RAI according to the present modification to the gNB **200**.

Alternatively, the UE **100** may include, in the RAI, an indicator (flag) for explicitly indicating whether or not the UE **100** supports reception of the conditional RRC release message.

Modification 2 of First Embodiment

(93) Next, a description will be given of differences between modification 2 of the first embodiment, and the first embodiment and modification 1 thereof described above. FIG. **9** is a diagram illustrating operations of the mobile communication system according to modification 2 of the first embodiment.

(94) As illustrated in FIG. **9**, steps S**111** to S**113** are the same as the first embodiment described above.

(95) In the present modification, after receiving a conditional RRC release message from the gNB

200, the UE 100 transmits, to the gNB 200, a cancel request to cancel the releasing or suspending of the RRC connection (step S114), or receives a cancel request from the gNB 200 (step S115). A response to the cancel request may be further transmitted and received. Such signaling may be made by way of an RRC message or a MAC Control Element (CE). The cancel request may include an information element indicating a reason for cancellation.

(96) Thus, even after the conditional RRC release message has been received, the RRC connected mode can be continued to be maintained, such as in a case where data is unexpectedly generated.

(97) Here, a specific example of such cancellation will be described.

(98) Firstly, one example of a case in which the UE 100 initiates the cancellation is a case in which the UE 100 anticipates generation of new uplink data. In this case, the UE 100 transmits a cancel request to the gNB 200 (step S114). The cancel request may be an RAI cancel request that requests cancellation of the RAI. When the gNB 200 receives the cancel request from the UE 100, the gNB 200 determines whether or not to cancel RRC connection release or suspension by the conditional RRC release message. Then, when the gNB 200 determines to cancel the RRC connection release or suspension, the gNB 200 may transmit a conditional RRC Release cancel command for indicating the cancellation to the UE 100.

(99) Secondly, one example of case in which the gNB 200 initiates the cancellation is a case in which the gNB 200 anticipates generation of new downlink data. For example, when the AMF 300 transmits notification about new downlink data to the gNB 200, the gNB 200 can anticipate generation of new downlink data. In this case, the gNB 200 transmits the above-described cancel request or conditional RRC Release cancel command to the UE 100.

(100) The UE 100 that has received the conditional RRC Release cancel command discards a conditional RRC Release command. Specifically, the UE 100 discards condition information configured and stored with the conditional RRC release message. When the UE 100 has performed measurement for evaluating whether or not the condition has been met (for example, measurement of RSRP) and timer operation, the UE 100 stops the measurement and the timer operation. Then, the UE 100 maintains the RRC connected mode.

(101) Alternatively, instead of discarding the conditional RRC Release command, the UE 100 may store condition information corresponding to the conditional RRC Release command, and suspend evaluation as to whether or not the condition has been met. Then, the UE 100 may restart application of the stored condition information when resumption of the conditional RRC Release command is indicated by the gNB 200.

Second Embodiment

(102) Next, a difference between a second embodiment and the first embodiment described above will be described. FIG. 10 is a diagram illustrating operations of the mobile communication system according to the second embodiment.

(103) As illustrated in FIG. 10, in step S201, the UE 100 is in the RRC connected mode in a cell of the gNB 200. The UE 100 in the RRC connected mode performs data communication with the gNB 200.

(104) In step S202, the UE 100 transmits uplink data to the gNB 200 via the PUSCH, or receives downlink data from the gNB 200 via the PDSCH.

(105) In step S203, the UE 100 and the gNB 200 complete the data communication. Completion of the data communication refers to uplink data transmission completion when only uplink data transmission is performed, refers to downlink data transmission completion when only downlink data transmission is performed, and refers to uplink and downlink data transmission completion when uplink and downlink data transmission is performed.

(106) In step S204, the UE 100 may transmit, to the gNB 200, a RAI which is an indicator indicating that data to be transmitted to the gNB 200 will not occur in the near future. The RAI may be a buffer status report indicating that a buffer size value is zero. The RAI may be enabled only when RAI activation is configured from the gNB 200. However, step S4 is not necessarily required

and step **S205** may be performed without step **S204**.

(107) In step **S205**, the gNB **200** transmits, to the UE **100**, an RRC release message with a condition (predetermined condition) for releasing the RRC connection of the UE **100**. The RRC release message according to the second embodiment is an RRC release message which makes the UE **100** transition to the RRC inactive mode, and which is associated with the predetermined condition that makes the UE **100** transition to the RRC idle mode.

(108) In the second embodiment, the predetermined condition is a condition that a predetermined time elapses from the reception of the RRC release message or the transition to the RRC inactive mode. The predetermined condition according to the second embodiment may be a condition that a trigger for transitioning to the RRC connected mode does not occur within the predetermined time after the UE **100** transitions to the RRC inactive mode.

(109) Such a conditional RRC release message may be an RRC release message including condition information specifying the predetermined condition. The conditional RRC release message can be a new RRC release message having a different format from the general RRC release message.

(110) In the second embodiment, the conditional RRC release message may include the SuspendConfig described above. The conditional RRC release message may also include at least one parameter to adopt after transitioning to the RRC idle mode.

(111) In step **S206**, the UE **100** transitions to the RRC inactive mode in response to receiving the conditional RRC release message from the gNB **200**. The UE **100** starts a timer corresponding to the predetermined time when receiving the conditional RRC release message or transitioning to the RRC inactive mode.

(112) The predetermined time (timer value) may be configured for the UE **100** as the configuration information (information element) in the conditional RRC release message, may be configured for the UE **100** as the configuration information (information element) in a system information block (SIB) broadcast by the gNB **200**, or a timer value defined by communication standards may be configured for the UE **100** in advance at the time of shipping the UE **100**.

(113) The UE **100** may stop the timer in a case where a trigger for transitioning to the RRC connected mode occurs while the timer is running. The trigger for transitioning to the RRC connected mode may be the reception by the UE **100** of a paging message from the gNB **200**, or the occurrence, in the UE **100**, of uplink data to be transmitted. Alternatively, the UE **100** may stop the timer when transitioning to the RRC connected mode after the trigger for transitioning to the RRC connected mode occurs while the timer is running.

(114) The UE **100** may reset the timer value and restart the timer in a case of receiving an indication of timer restart from the gNB **200** while the timer is running. The indication may be included in a RAN paging message transmitted by the eNB **200** or may be included in the system information. Thus, the gNB **200** can control the time for which the UE **100** is maintained in the RRC inactive mode.

(115) Here, the description proceeds under the assumption that the UE **100** has allowed the timer to expire without stopping the timer.

(116) In step **S207**, the UE **100** autonomously transitions to the RRC idle mode from the RRC inactive mode in response to the timer expiring.

(117) On the other hand, the gNB **200** also manages the timer in the same manner as the UE **100**, and assumes that the UE **100** has transitioned to the RRC idle mode in response to the expiration of the timer. Then, in step **S208**, the gNB **200** notifies the core network (AMF **300**) of the UE **100** having transitioned to the RRC idle mode and the identifier of the UE **100**. The notification may be performed on the NG interface.

(118) Note that, instead of the timer, the predetermined time may be represented by the number of paging reception occasions (number of times) during the RRC inactive mode. The paging reception occasions occur in a cycle corresponding to ran-PagingCycle in SuspendConfig. For example, the

UE **100** may transition from the RRC inactive mode to the RRC idle mode in a case of receiving no paging message in a configured number of paging reception occasions.

(119) According to the second embodiment, even in the case where the gNB **200** does not include sufficient data to determine whether to make the UE **100** transition to the RRC idle mode or the RRC inactive mode, the UE **100** autonomously transitions to the RRC idle mode from the RRC inactive mode, and can thereby use the RRC idle mode and the RRC inactive mode appropriately for different purposes.

Other Embodiments

(120) FIG. **11** is a diagram illustrating operations according to other embodiments. The present communication control method is a method in view of carrier aggregation, in which a primary cell and one or more secondary cells are used simultaneously.

(121) As illustrated in FIG. **11**, in step **S301**, the UE **100** predicts the amount of uplink data generated in a certain future period. For example, the UE **100** predicts the amount of uplink data generated in a certain future period, based on information obtained in an upper layer with respect to a radio layer (a PHY layer, a MAC layer, an RLC layer, a PDCP layer, and an SDAP layer), for example, an application layer or the like. The UE **100** may perform prediction based on a type of an application, or may perform prediction based on the number of applications to be executed.

(122) In step **S302**, the UE **100** identifies the amount of uplink data already stored in an uplink data buffer of the UE **100**. For example, the UE **100** identifies the amount of uplink data stored in the uplink data buffer of all of or a part of the layers of the MAC layer, the RLC layer, and the PDCP layer. In other words, the UE **100** identifies the amount of uplink data (specifically, a buffer status report (BSR) report value) to be reported with a BSR transmitted from the UE **100** to the gNB **200** in the MAC layer. Note that step **S302** may be performed before step **S301**.

(123) In step **S302**, the UE **100** notifies the gNB **200** of the amount of uplink data, which is obtained by subtracting the amount of uplink data identified in step **S302** from the amount of uplink data predicted in step **S301**. Step **S3** may be performed at the time of message 3 (Msg3) transmission or message 5 (Msg5) transmission in the random access procedure, or at the time of subsequent message transmission.

(124) The amount of uplink data notified to the gNB **200** in step **S302** is used for control related to the secondary cell configured for the UE **100** by the gNB **200**. For example, the control related to the secondary cell may include determination as to whether the secondary cell is configured for the UE **100** or the configuration is released, or may include determination as to whether the secondary cell already configured for the UE **100** is enabled (activated) or disabled (deactivated).

(125) In this manner, instead of directly notifying a prediction value of the amount of uplink data generated in a certain future period to the gNB **200**, the UE **100** notifies a result obtained by subtracting the BSR report value from the prediction value to the gNB **200**. Here, the gNB **200** can identify the BSR report value, based on the BSR transmitted from the UE **100**.

(126) When the BSR report value is included in the prediction value of the amount of uplink data generated in a certain future period, the amount of uplink data corresponding to the BSR report value may be notified twice, which may cause the gNB **200** to estimate the amount of uplink data of the UE **100** to be larger than the actual amount, and thereby configure or enable the secondary cell more than necessary, thus increasing the power consumption of the UE **100**. According to the method illustrated in FIG. **11**, the overlapping part as described above can be eliminated, and thus by the gNB **200** more appropriately performing the control related to the secondary cell, the power consumption of the UE **100** can be reduced.

(127) Note that the method illustrated in FIG. **11** may be applied after the UE **100** transmits the BSR to the gNB **200**. Specifically, when the UE **100** does not transmit the BSR to the gNB **200**, the prediction value of the amount of uplink data generated in a certain future period may be directly notified to the gNB **200**.

(128) The embodiments above are described mainly for a 5G system (NR), but the operations

according to the embodiments may be applied to LTE.

(129) In the embodiments described above, the UE **100** may be a UE for a machine type communication application or an IoT application.

(130) Note that a program causing a computer to execute each of the processes performed by the UE **100** or the gNB **200** may be provided. The program may be recorded in a computer readable medium. Use of a computer readable medium enables the program to be installed on a computer. Here, the computer readable medium on which the program is recorded may be a non-transitory recording medium. The non-transitory recording medium is not particularly limited, and may be, for example, a recording medium such as a CD-ROM, a DVD-ROM, or the like.

(131) Circuits for performing the processes performed by the UE **100** or the gNB **200** may be integrated to configure at least a portion of the UE **100** or the gNB **200** as a semiconductor integrated circuit (chipset, SoC).

(132) An embodiment has been described above in detail with reference to the drawings; however, specific configurations are not limited to those described above, and various design modifications can be made without departing from the gist of the present disclosure.

Claims

1. A communication control method comprising: transmitting, by a user equipment to a network node, first information indicating that the user equipment prefers one of a RRC idle mode and a RRC inactive mode when data to be transmitted to the network node is not to occur in near future; and in response to changing a preferred RRC mode from a RRC mode indicated by the first information to a RRC connected mode, transmitting, by the user equipment to the network node, second information indicating that the user equipment prefers to revert the first information to leave the RRC connected mode.
2. A user equipment comprising: a controller configured to: transmit, to a network node, first information indicating that the user equipment prefers one of a RRC idle mode and a RRC inactive mode when data to be transmitted to the network node is not to occur in near future; and in response to changing a preferred RRC mode from a RRC mode indicated by the first information to a RRC connected mode, transmit, to the network node, second information indicating that the user equipment prefers to revert the first information to leave the RRC connected mode.
3. A chipset for a user equipment, the chipset configured to execute processing of: transmitting, to a network node, first information indicating that the user equipment prefers one of a RRC idle mode and a RRC inactive mode when data to be transmitted to the network node is not to occur in near future; and in response to changing a preferred RRC mode from a RRC mode indicated by the first information to a RRC connected mode, transmitting, to the network node, second information indicating that the user equipment prefers to revert the first information to leave the RRC connected mode.
4. A non-transitory computer-readable medium comprising, stored thereupon, computer program instructions for execution by a user equipment, the program instructions being configured to cause the user equipment to execute processing of: transmitting, to a network node, first information indicating that the user equipment prefers one of a RRC idle mode and a RRC inactive mode when data to be transmitted to the network node is not to occur in near future; and in response to changing a preferred RRC mode from a RRC mode indicated by the first information to a RRC connected mode, transmitting, to the network node, second information indicating that the user equipment prefers to revert the first information to leave the RRC connected mode.
5. A system comprising: a user equipment configured to: transmit, to a network node, first information indicating that the user equipment prefers one of a RRC idle mode and a RRC inactive mode when data to be transmitted to the network node is not to occur in near future; and in response to changing a preferred RRC mode from a RRC mode indicated by the first information to

a RRC connected mode, transmit, to the network node, second information indicating that the user equipment prefers to revert the first information to leave the RRC connected mode.
